Linear#

https://pytorch.org/docs/stable/generated/torch.ao.nn.quantized.Linear.html

Description:

A quantized linear module with quantized tensor as inputs and outputs. We adopt the same interface as torch.nn.Linear, please see https://pytorch.org/docs/stable/nn.html#torch.nn.Linear for documentation. Similar to Linear, attributes will be randomly initialized at module creation time and will be overwritten later weight (Tensor) – the non-learnable quantized weights of the module of shape (out_features,in_features)(\text{out_features}, \text{in_features}) (out_features,in_features). bias (Tensor) – the non-learnable bias of the module of shape (out_features)(\text{out_features})(out_features). If bias is True, the values are initialized to zero. scale – scale parameter of output Quantized Tensor, type: double

Function Signature

```
class torch.ao.nn.quantized.Linear(in_features,
out_features, bias_=True, dtype=torch.qint8)[source]#

Variables

classmethod from_float(mod,
use_precomputed_fake_quant=False)[source]#

Parameters

classmethod from_reference(ref_qlinear, output_scale,
output_zero_point)[source]#
```

Code Examples

```
>>> m = nn.quantized.Linear(20, 30)
>>> input = torch.randn(128, 20)
>>> input = torch.quantize_per_tensor(input, 1.0, 0,
torch.quint8)
>>> output = m(input)
>>> print(output.size())
torch.Size([128, 30])
```

Detailed Documentation

Documentation

A quantized linear module with quantized tensor as inputs and outputs. We adopt the same interface as torch.nn.Linear, please see https://pytorch.org/docs/stable/nn.html#torch.nn.Linear for documentation.

Similar to Linear, attributes will be randomly initialized at module creation time and will be overwritten later

weight (Tensor) – the non-learnable quantized weights of the module of shape (out_features,in_features)(\text{out_features}, \text{in_features})(out_features,in_features).

bias (Tensor) – the non-learnable bias of the module of shape (out_features)(\text{out_features})(out_features). If bias is True, the values are initialized to zero.

scale – scale parameter of output Quantized Tensor, type: double

zero_point – zero_point parameter for output Quantized Tensor, type: long

Create a quantized module from an observed float module

mod (Module) – a float module, either produced by torch.ao.quantization utilities or provided by the user

use_precomputed_fake_quant (bool) – if True, the module will reuse min/max values from the precomputed fake quant module.

Create a (fbgemm/qnnpack) quantized module from a reference quantized module

ref_qlinear (Module) – a reference quantized linear module, either produced by torch.ao.quantization utilities or provided by the user

output_scale (float) – scale for output Tensor output_zero_point (int) – zero point for output Tensor